

Practice question

PCA in Face recognition in biometrics

Consider the following four 2×2 training face images.

$$F_1 = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$$

$$F_2 = \begin{bmatrix} 3 & 2 \\ 4 & 3 \end{bmatrix}$$

$$F_3 = \begin{bmatrix} 4 & 3 \\ 5 & 4 \end{bmatrix}$$

$$F_4 = \begin{bmatrix} 5 & 4 \\ 6 & 5 \end{bmatrix}$$

After applying PCA, the following Eigenfaces are selected.

Largest Eigenvalues

$$[20, 3]$$

Eigenface 1

$$v_1 = \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix}$$

Eigenface 2

$$v_2 = \begin{bmatrix} 0.5 \\ -0.5 \\ 0.5 \\ -0.5 \end{bmatrix}$$

Answer the following:

1. Convert each image into a face vector.
2. Compute the normalized vectors.
3. Compute the weight vectors for all four faces.
4. A new face (testing face is) :

$\begin{bmatrix} 3 & 2 \\ 4 & 3 \end{bmatrix}$ is presented. Recognize the person using Euclidean distance.